Date	03 October 2026
Team ID	SWTID-2026-6413
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

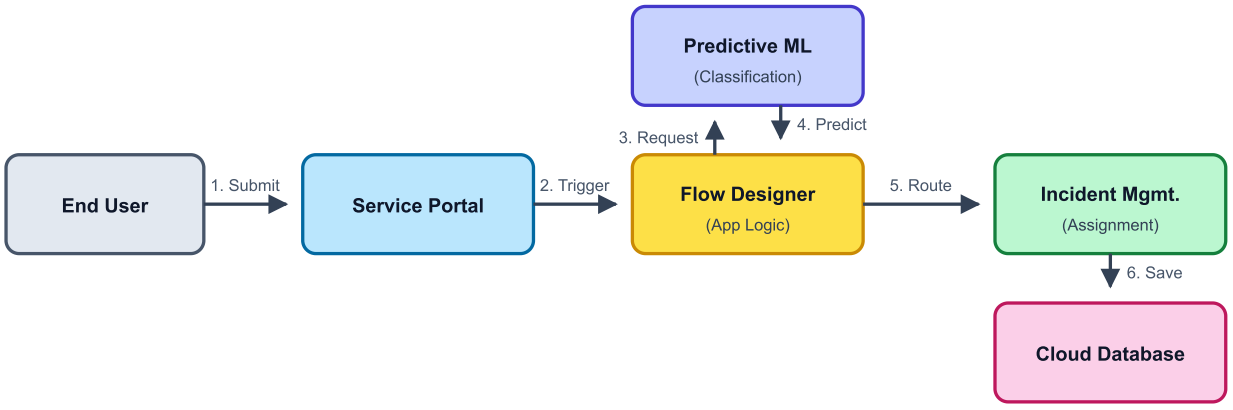
Project Design Phase-II

Technology Stack (Architecture & Stack)

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table 1 & table 2.

Example: Order processing during pandemics for offline mode
Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

S.No	Component	Description	Technology
1.	**User Interface**	How users interact with the application to submit or track tickets (e.g., Service Portal, Virtual Agent)	HTML, CSS, AngularJS, React JS

S.No	Component	Description	Technology
4.	**Application Logic-3**	Core ITSM logic for incident creation, updates, and SLA tracking	JavaScript (Server-side Business Rules)
5.	**Database**	Relational database storing ticket records, CMDB, and user data	MariaDB / MySQL
6.	**Cloud Database**	Managed database service on the cloud for scalable record management	PaaS Cloud Database
7.	**File Storage**	Storage requirements for user-uploaded attachments, screenshots, and logs	Cloud Blob Storage / PaaS Attachment Architecture
8.	**External API-1**	Enterprise messaging API for automated SMS or platform notifications	Twilio API / Slack API
9.	**External API-2**	Authentication and Single Sign-On (SSO) integration for user access	Azure Active Directory (Azure AD) / Okta API
10.	**Machine Learning Model**	Classification model to categorize, prioritize, and assign tickets based on historical text data	Classification Framework / Predictive Intelligence
11.	**Infrastructure (Server / Cloud)**	Application deployment on a scalable cloud environment Local Server Configuration: NA Cloud Server Configuration: Dedicated SaaS Instances	Enterprise SaaS Cloud Platform (e.g., ServiceNow)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	**Open-Source Frameworks**	Frontend and scripting frameworks utilized for custom widgets and portals	AngularJS, Bootstrap
2.	**Security Implementations**	Role-based access control, encryption in transit/rest, and secure API communication	ACLs (Access Control Lists), OAuth 2.0, TLS 1.2+
3.	**Scalable Architecture**	Multi-instance architecture designed to handle varying volumes of incoming IT support tickets	Cloud-native SaaS Architecture
4.	**Availability**	High availability setup ensuring continuous uptime for the ITSM platform	Active-Active Datacenter pairing, Advanced Load Balancers
5.	**Performance**	Optimized processing times for real-time automated ticket routing	Edge Caching, Database Indexing, CDN

References:

- https://c4model.com/
- https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/
- https://www.ibm.com/cloud/architecture
- https://aws.amazon.com/architecture
- https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d